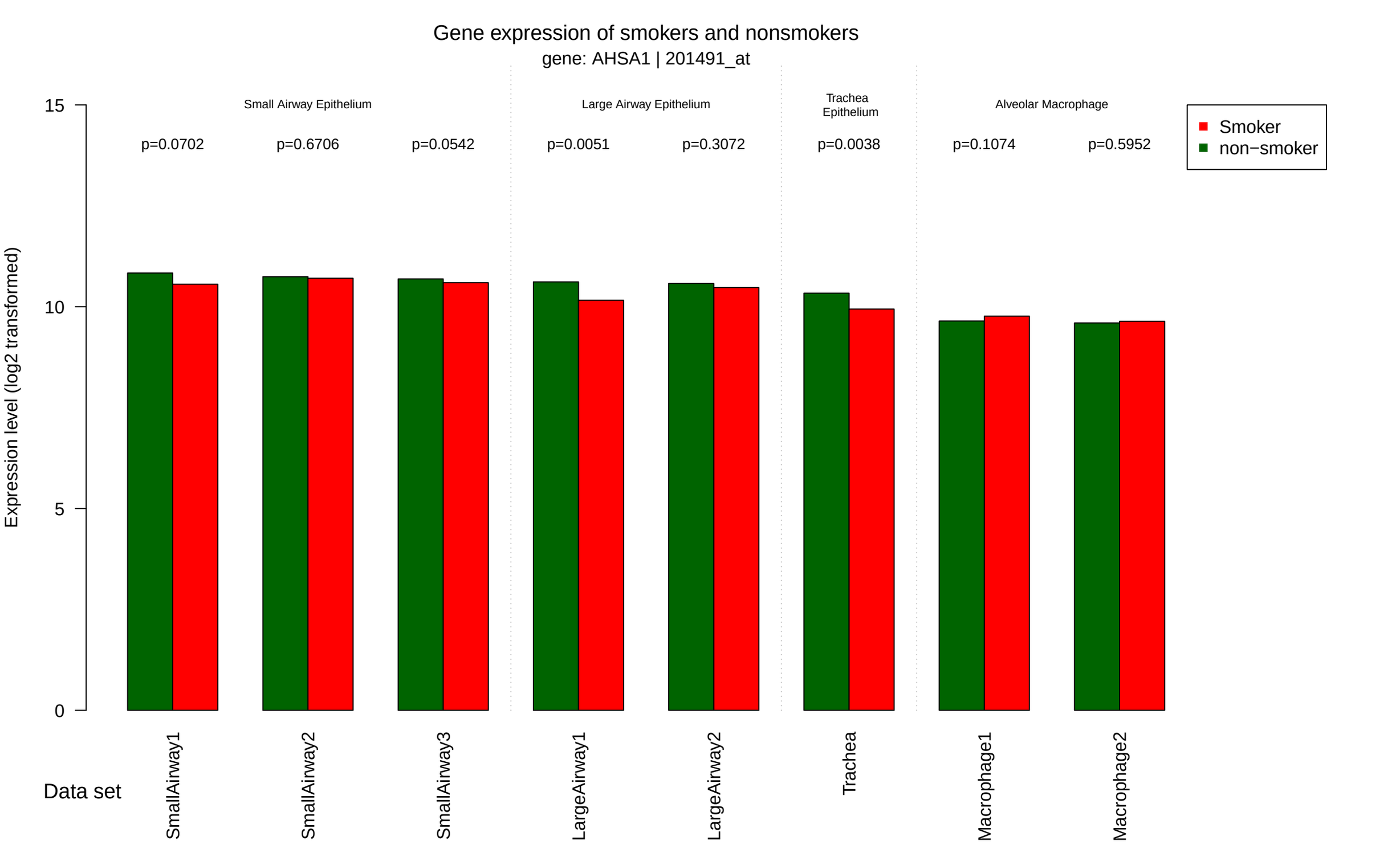
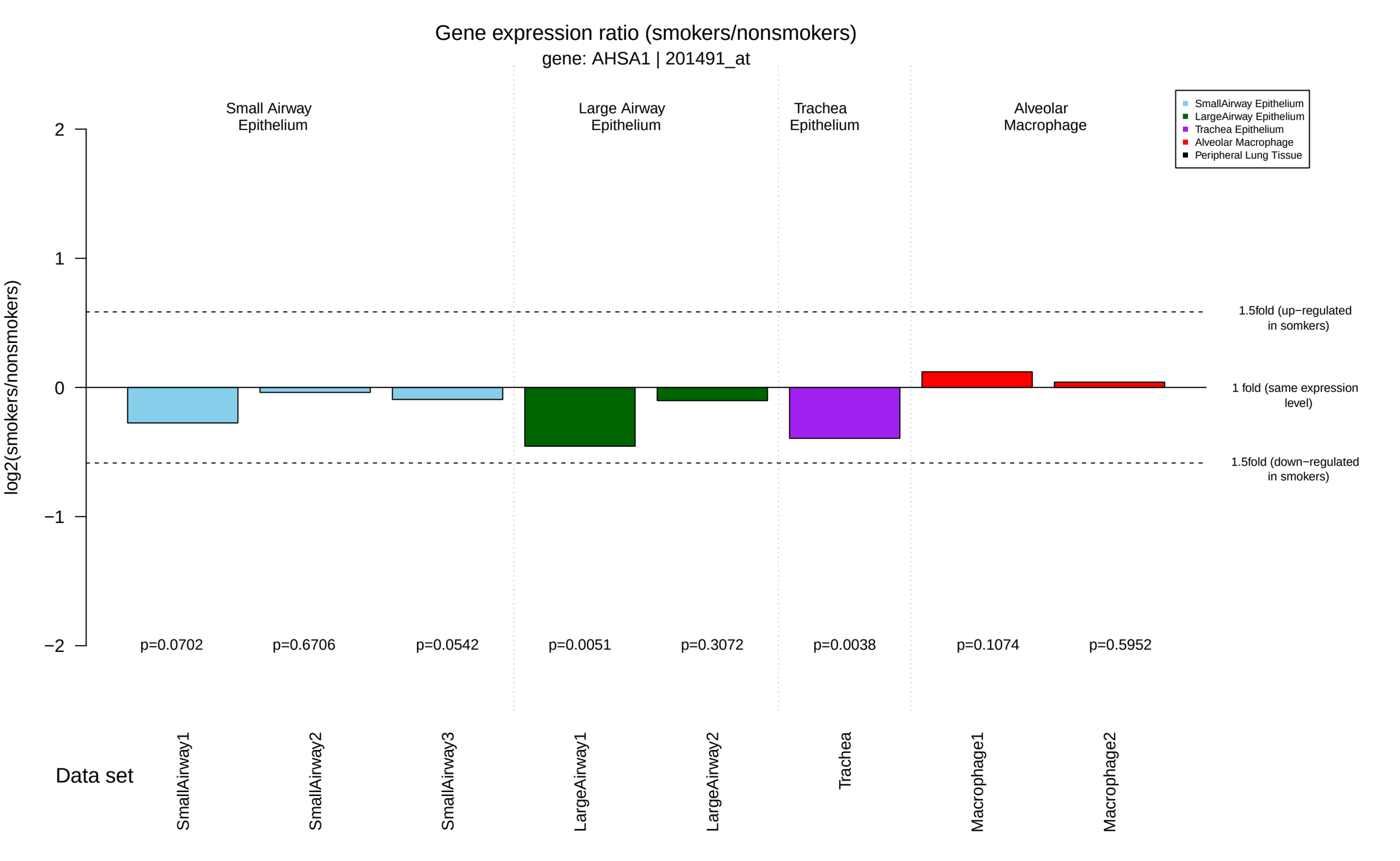
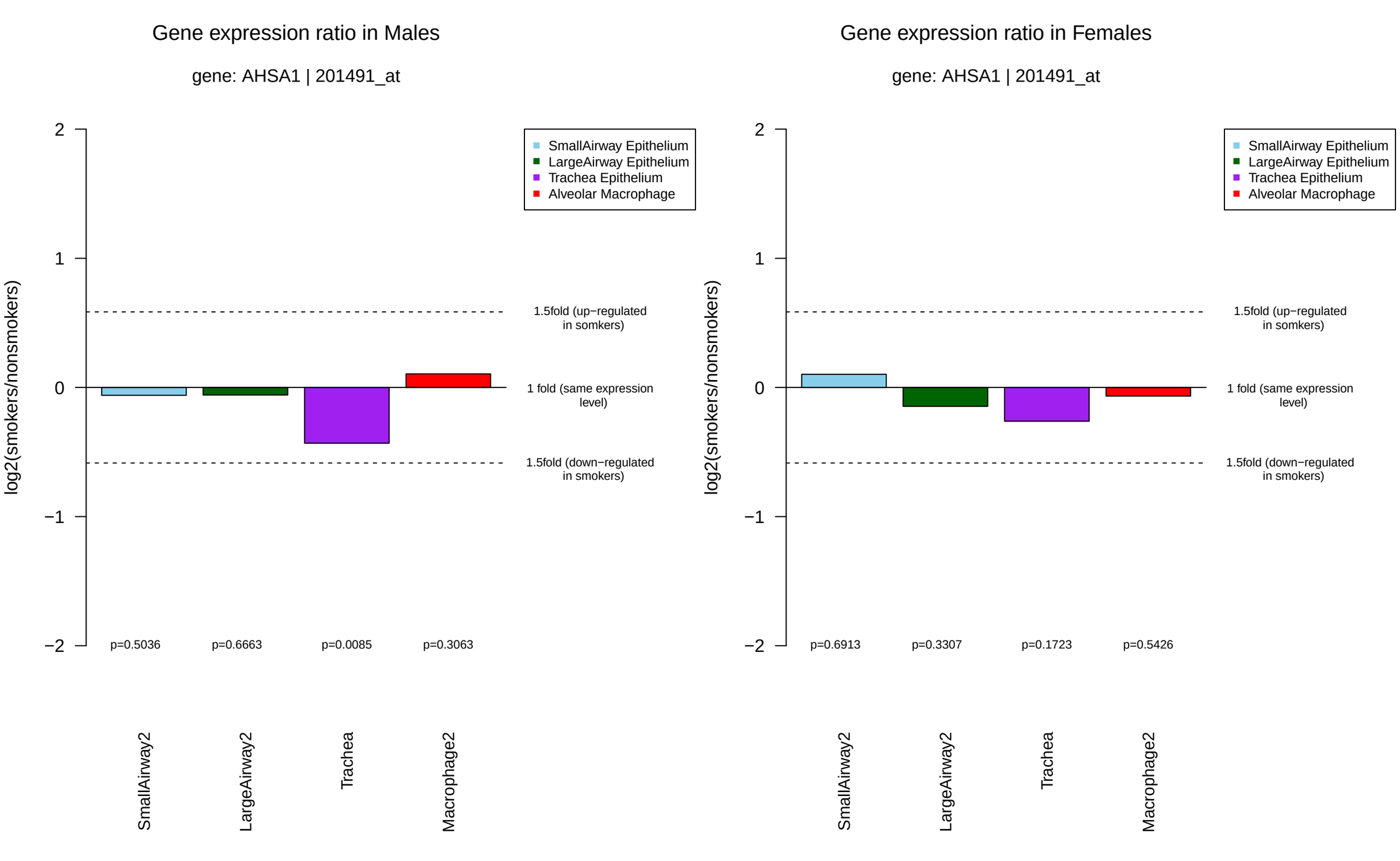
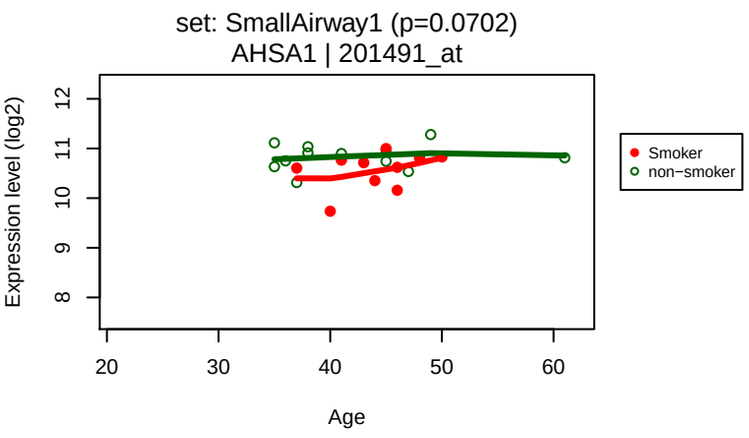
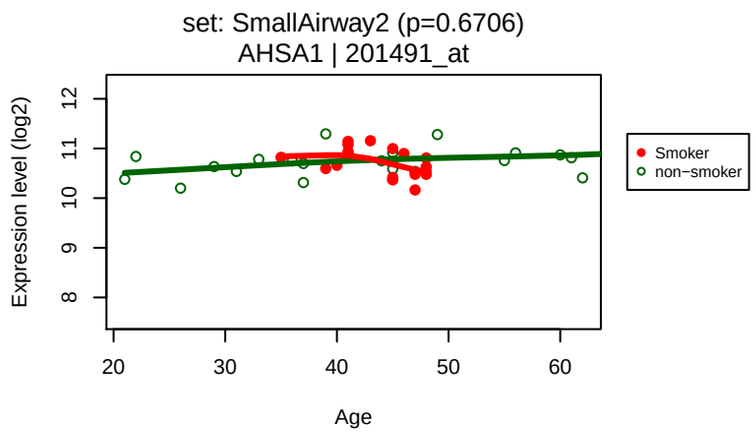
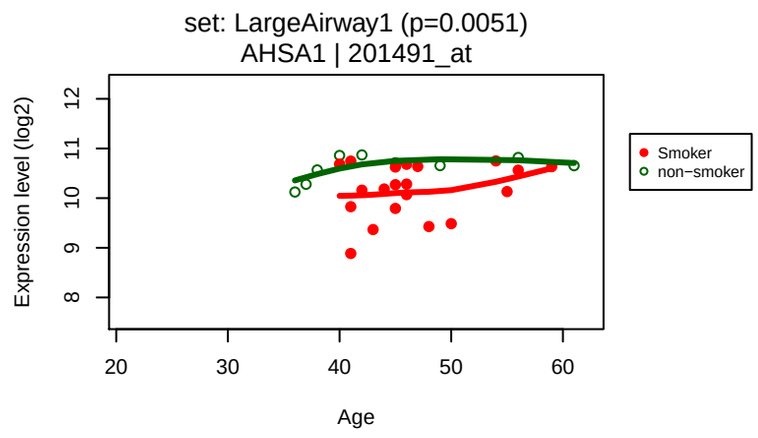

Gene expression of smokers and nonsmokers
gene: AHSA1 | 201491_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0702
p=0.6706
p=0.0542
p=0.0051
p=0.3072
p=0.0038
p=0.1074
p=0.5952
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

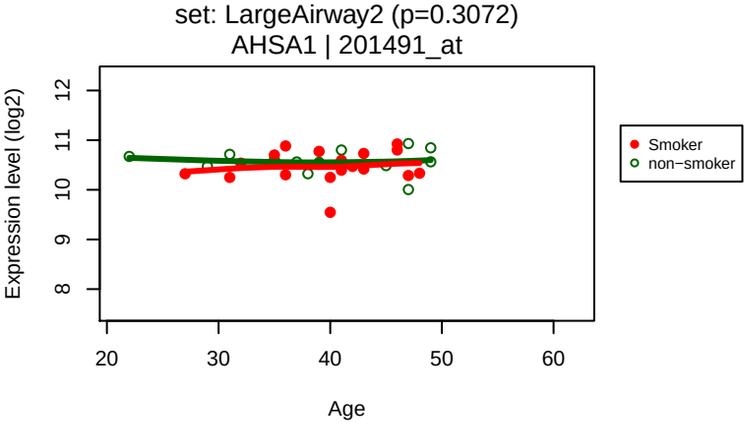

Gene expression ratio (smokers/nonsmokers)
gene: AHSA1 | 201491_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0702
p=0.6706
p=0.0542
p=0.0051
p=0.3072
p=0.0038
p=0.1074
p=0.5952
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

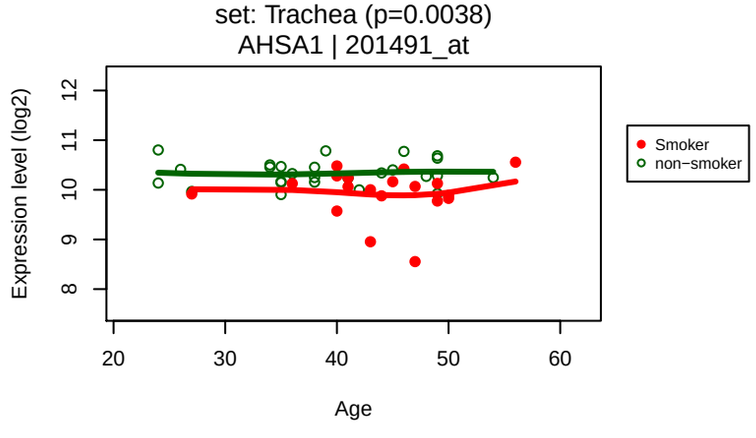

Gene expression ratio in Males
gene: AHSA1 | 201491_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5036
p=0.6663
p=0.0085
p=0.3063
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AHSA1 | 201491_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6913
p=0.3307
p=0.1723
p=0.5426
SmallAirway2
LargeAirway2
Trachea
Macrophage2

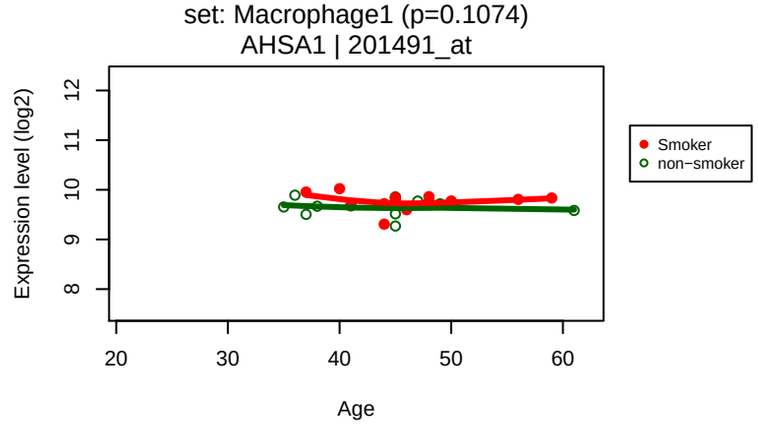

set: SmallAirway1 (p=0.0702)
AHSA1 | 201491_at
Expression level (log2)
Age
Smoker
non−smoker

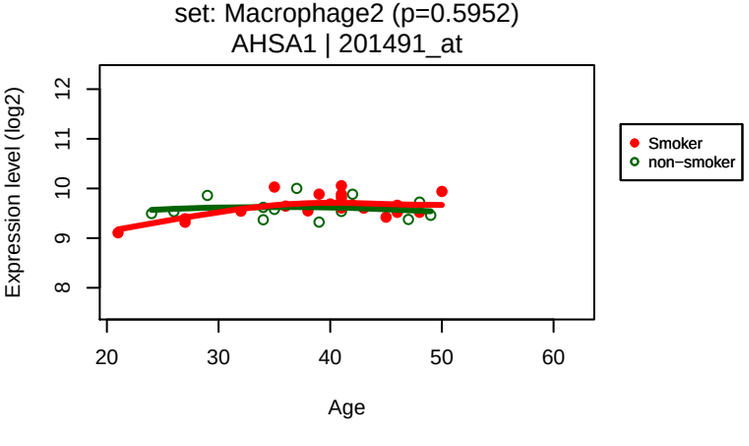

set: SmallAirway2 (p=0.6706)
AHSA1 | 201491_at
Expression level (log2)
Age
Smoker
non−smoker

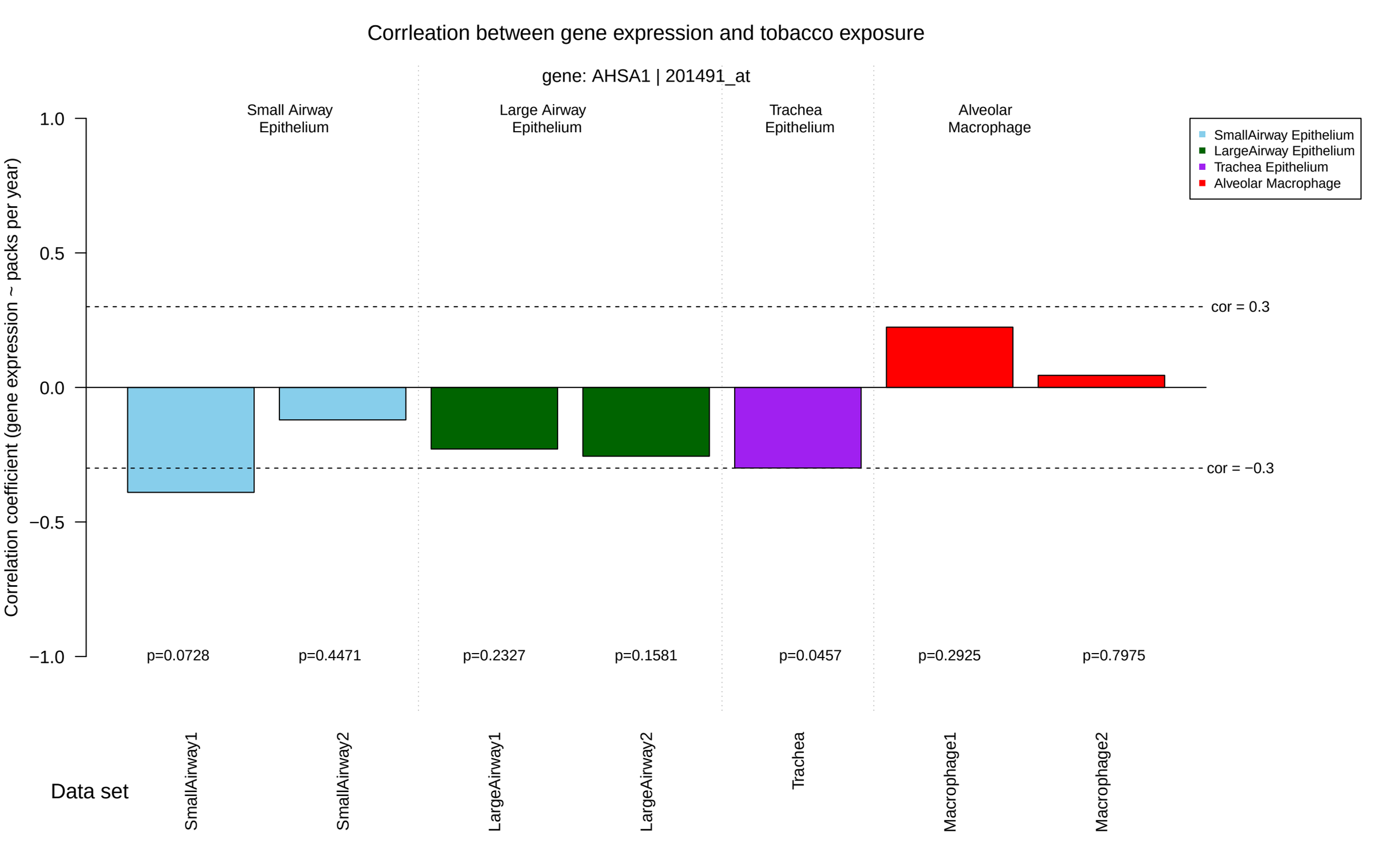

set: LargeAirway1 (p=0.0051)
AHSA1 | 201491_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3072)
AHSA1 | 201491_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0038)
AHSA1 | 201491_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1074)
AHSA1 | 201491_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5952)
AHSA1 | 201491_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AHSA1 | 201491_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0728
p=0.4471
p=0.2327
p=0.1581
p=0.0457
p=0.2925
p=0.7975
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2